PhyGRU Application: Vehicle Dynamics Modelling

Girolamo Oddo

March 26, 2026

1 Introduction

This work investigates the application of the PhyGRU architecture to vehicle dynamics modeling. A comparison is carried out between a vanilla GRU and the PhyGRU for sequence-to-sequence prediction of vehicle states from driver control inputs. The objective is to estimate the signals

$$\mathbf{y} = [a_x,\ a_y,\ r]^T \tag{1}$$

from the control input sequence

$$\mathbf{u} = [\delta,\ \text{throttle},\ \text{brake}]^T, \tag{2}$$

where a_x and a_y denote longitudinal and lateral accelerations, r denotes yaw rate and δ the steering input. The GRU is used as a purely data-driven baseline, whereas the PhyGRU incorporates a bicycle-model prior.

2 Dataset generation and vehicle models

The data are generated synthetically by a nonlinear 7-DOF vehicle simulator. The full simulator state is

$$\mathbf{x} = [v_x,\ v_y,\ r,\ \omega_{fl},\ \omega_{fr},\ \omega_{rl},\ \omega_{rr}]^T, \tag{3}$$

with longitudinal velocity v_x, lateral velocity v_y, yaw rate r, and the four wheel speeds. Smooth random control signals are applied to the simulator to obtain training, validation, and test sequences of equal structure, only time length changes, with the following duration: training 10 s; validation 10 s; test 50 s. The same procedure is used for all splits, while Gaussian measurement noise is added only to the output signals used for training and evaluation.
The simulator computes slip angles as

$$\alpha_f = \delta - \arctan 2(v_y + l_f r,\ v_x), \tag{4}$$
$$\alpha_r = -\arctan 2(v_y - l_r r,\ v_x), \tag{5}$$

and lateral tire forces as

$$F_{yf} = 2C_f \tanh(\alpha_f), \tag{6}$$
$$F_{yr} = 2C_r \tanh(\alpha_r). \tag{7}$$

The planar body dynamics are then given by

$$\dot{v}_x = \frac{F_x - F_{yf}\sin(\delta) + m v_y r}{m}, \tag{8}$$
$$\dot{v}_y = \frac{F_{yf}\cos(\delta) + F_{yr} - m v_x r}{m}, \tag{9}$$
$$\dot{r} = \frac{l_f F_{yf}\cos(\delta) - l_r F_{yr}}{I_z}, \tag{10}$$

with

$$F_x = F_{x,\max}\,\text{throttle} - F_{b,\max}\,\text{brake} - C_d v_x|v_x| - C_{rr}\tanh(v_x). \tag{11}$$

The output signals are defined as

$$a_x = \dot{v}_x - v_y r, \qquad a_y = \dot{v}_y + v_x r. \tag{12}$$

A reduced bicycle model is used inside the PhyGRU as prior knowledge. Its state is

$$\mathbf{s} = [v_x,\ v_y,\ r]^T, \tag{13}$$

and it uses the same input vector $\mathbf{u}$. The bicycle model preserves the planar rigid-body dynamics of the vehicle while neglecting the wheel-speed dynamics and other higher-order effects. As a consequence, it provides a compact physics-based approximation of the same output variables,

$$\mathbf{y}_{phy} = [a_x,\ a_y,\ r]^T, \tag{14}$$

while remaining intentionally simpler than the 7-DOF ground-truth generator.

3 Learning setup

The baseline GRU is a vanilla recurrent model that directly maps the input sequence to the output sequence. The PhyGRU instead exploit the bicycle-model as physical prior during learning. At each time step, the PhyGRU cell takes the control input and the current state as input. The bicycle-model prior produces a physics-based state evolution, which is combined with a learned latent dynamics branch through a gated recurrent update; the resulting state is then used to generate a physics-informed output, which is refined by three independent residual heads, one for each predicted channel. This design preserves physical consistency while allowing the model to learn the mismatch between the simplified bicycle equations and the full 7-DOF dynamics.

Before training, the outputs are standardized using the statistics of the training set, whereas the inputs remain in raw physical units. The same preprocessing is applied to the validation and test sets. Both models are trained with mean squared error loss and the Adam optimizer. The training loop has been designed to bring the two models to comparable performance on validation, so as to better assess later the generalization on test set with longer time series. Training was stopped once both models reached a comparable target validation MSE, ensuring a fair comparison of their generalization capabilities.

Model	Epoch	Train MSE	Val MSE
GRU	120	2.79×10^{-2}	2.81×10^{-2}
PhyGRU	10	2.91×10^{-2}	2.92×10^{-2}

Table 1: Training and validation MSE at selected epochs.

4 Evaluation protocol & Results

The test set is generated with the same synthetic procedure used for training and validation, so that the only difference lies in the held-out control sequences. Performance is evaluated on the noisy output signals through mean squared error and per-channel error over all testing trajectories.

The inference average time per-step are respectively of 0.0044 ms for GRU and 0.0298 ms for PhyGRU on Intel(R) Xeon(R) CPU @ 2.20 GHz.

Model	a_x	a_y	yaw rate
GRU	4.86×10^{-2}	6.10×10^{0}	6.78×10^{-4}
PhyGRU	2.33×10^{-2}	1.29×10^{-1}	2.56×10^{-4}

Table 2: Test MSE per output component.

In addition, a qualitative trajectory reconstruction is performed by double integrating the predicted signals, in order to inspect the accumulated error in planar motion. The final result reported here is the grid of 36 reconstructed trajectories from the test set of 50 s duration. Each subplot shows the trajectory evolution for one test sequence, allowing a compact visual comparison of the predicted motion across the full test distribution.

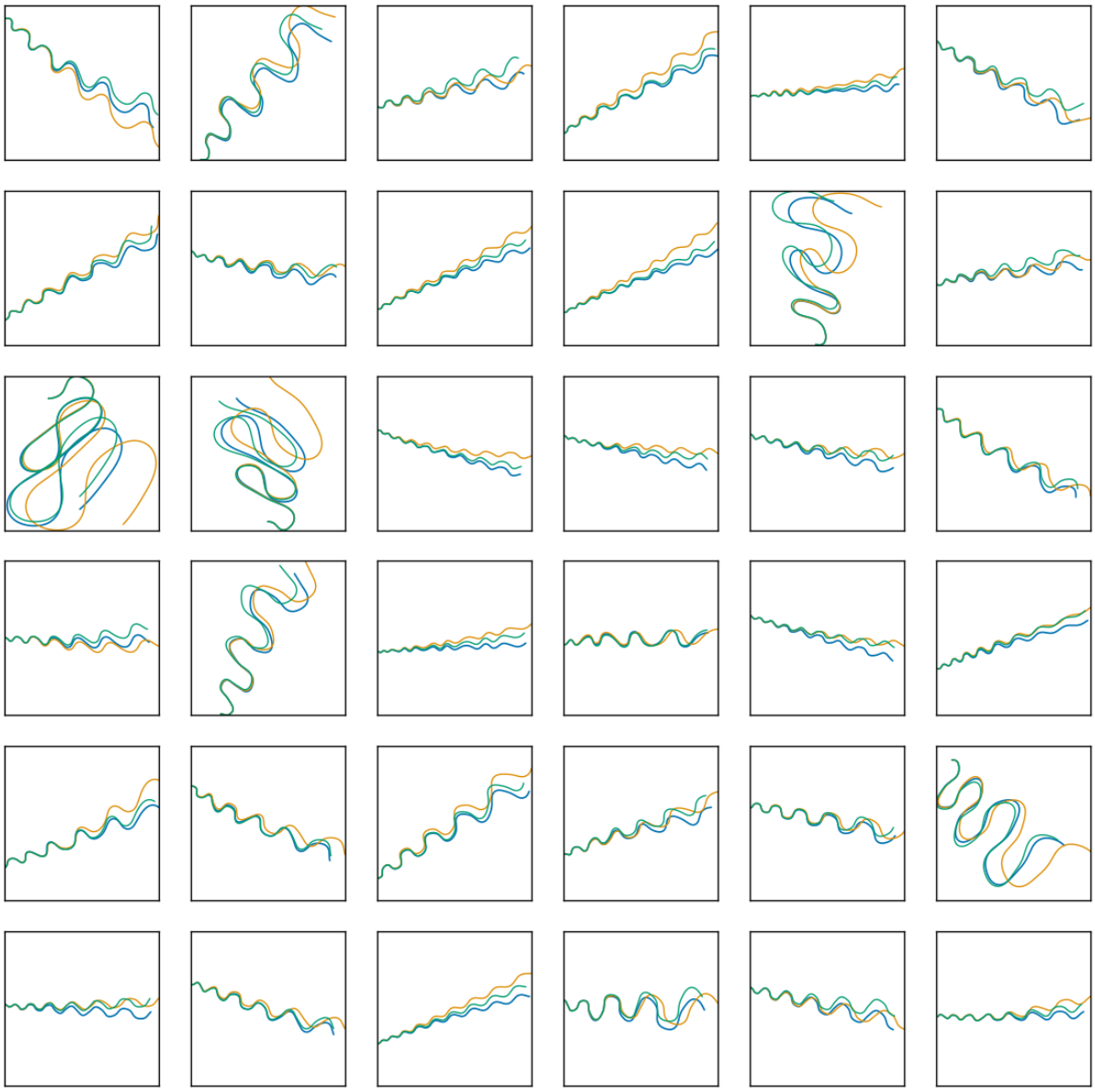

Figure 1: Test results, in blue the Ground Truth, in green the PhyGRU and in yellow the GRU.